

JUNGHWAN KIM

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RESEARCH INTEREST

Probabilistic Modeling, Variational Inference, Reinforcement Learning, Deep Learning, Representation Learning, Social Network Analysis, Recommender System

EDUCATION

Seoul National University, Seoul, Korea Aug. 2017
B.S. in Industrial Engineering and Mathematics (double major, *Cum Laude*)
Minor in Computer Science & Engineering

Korea Science Academy, Busan, Korea Feb. 2009
Major in Mathematics and Physics

RESEARCH AND WORK EXPERIENCE

Data Mining Laboratory Jan. 2016 - Present
Research Intern (Advisor: Prof. U Kang) *Seoul National University, Seoul, Korea*

- Research area: network representation learning, probabilistic modeling, and deep learning

WISEfn Corporation July 2012 - May 2015
Software Engineer *Seoul, Korea*

- Developed investment strategy evaluation software and workflow management system
- Managed database management tools and data integration procedures

Nonlinear Neurophysics Laboratory Mar. 2007 - Feb. 2008
Research Intern (Advisor: Prof. Soo Yong Kim, Kyungsik Kim) *KAIST, Daejeon, Korea*

- Research area: random matrix theory, network theory, and game theory

PUBLICATIONS

1. Haekyu Park, Hyunsik Jeon, Junghwan Kim, Beunguk Ahn, and U Kang, “UniWalk: Explainable and Accurate Recommendation for Rating and Network Data”, SIAM International Conference on Data Mining (SDM) 2018. (Under review)
2. Junghwan Kim, Haekyu Park, Ji-eun Lee, and U Kang, “SIDE: Representation Learning in Signed Directed Networks”, ACM International Conference on World Wide Web (WWW) 2018. (Under review)
3. Gyuchang Lim, SooYong Kim, Junghwan Kim, Pyungsoo Kim, Yoonjong Kang, Sanghoon Park, Inho Park, Sang-Bum Park, and Kyungsik Kim, “Structure of a financial cross-correlation matrix under attack”, Physica A: Statistical Mechanics and its Applications, 2009, 388.18: 3851-3858.

PATENTS

1. U Kang, Junghwan Kim, and Haekyu Park, “Apparatus and Method for Representation Learning in Signed Directed Networks”, Korean Patent 10-2017-0130914, 2017.
2. U Kang, Haekyu Park, Junghwan Kim, and Hyunsik Jeon, “Explainable and Accurate Recommender Method and System using Social Network Information and Rating Information”, Korean Patent 10-2017-0159167, 2017.

AWARDS AND HONORS

Bronze Medal, National Collegiate Mathematics Competition (for math major), *Korea Mathematical Society* 2015
Silver Medal, National Collegiate Mathematics Competition (for non-math major), *Korea Mathematical Society* 2011
National Scholarship for Science and Engineering, *Korea Student Aid Foundation* 2009-2015
Silver Medal, Korea Physics Olympiad, *Korea Physical Society* 2008
Merit-based Scholarship, *Kwangshin Petroleum Co., Kyobo Life Insurance Co., Chokwang Paint Ltd.* 2006-2008

INVITED TALKS

Invited talk, Emoney Inc. 2016
Title: Machine Learning and Deep Learning with Stock Price Data

TEACHING EXPERIENCE

Teaching Assistant, Basic Calculus 2 at Seoul National University Fall 2010
Teaching Assistant, Basic Physics 1 at Seoul National University Spring 2010

GRADUATE COURSEWORK

Probability Theory 1 Spring 2017
Theory of Statistics 1 Spring 2017
Topics in Applied Mathematics (Numerical Linear Algebra for Big Data) Spring 2017
Introduction to Computer Vision Spring 2017
Topics in Big Data Analytics (Advanced Social Network Analysis) Spring 2017
Probabilistic Graphical Model Fall 2016
Industrial Data Analytics Fall 2015

TECHNICAL STRENGTHS

Programming Languages	Python, C++, VBA (Advanced) Java, MATLAB, R, C# (Experienced)
Libraries	NumPy, SciPy, Pandas, TensorFlow (Advanced) PyTorch (Experienced)
Database Frameworks	Microsoft SQL (Advanced) MySQL, Oracle (Experienced)

REFERENCE

U Kang, Associate Professor

Department of Computer Science and Engineering

Seoul National University, Seoul, Korea

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